1. LLM Dual-Agent for Taste Preference Prediction

Accession number: 20261220329939
Authors: Yao, Kaifang (1)
Author affiliation: (1) Software Institute, Nanjing University, Nanjing, China
Corresponding author: Yao, Kaifang(978887223@qq.com)
Source title: 2025 4th International Conference on Cloud Computing, Big Data Application and Software Engineering, CBASE 2025
Abbreviated source title: Int. Conf. Cloud Comput., Big Data Appl. Softw. Eng., CBASE
Part number: 1 of 1
Issue title: 2025 4th International Conference on Cloud Computing, Big Data Application and Software Engineering, CBASE 2025
Issue date: 2025
Publication year: 2025
Pages: 456-461
Language: English
ISBN-13: 9798331564919
Document type: Conference article (CA)
Conference name: 2025 4th International Conference on Cloud Computing, Big Data Application and Software Engineering, CBASE 2025
Conference date: October 24, 2025 - October 26, 2025
Conference location: Hybrid, Chengdu, China
Conference code: 219700
Sponsor: Chengdu University of Technology; IEEE
Publisher: Institute of Electrical and Electronics Engineers Inc.
Abstract: Taste preference prediction plays a crucial role in enhancing consumer satisfaction and driving innovation in areas such as food science and personalized nutrition. Current approaches often rely on large language models (LLMs), yet these methods remain constrained by sparse domain knowledge, fragmented reasoning processes, and limited integration of heterogeneous user data. To address these challenges, we introduce AGENTPREDICT, an LLM-driven multi-agent framework that leverages advances in cloud computing and big data technologies and synergizes a vector retrieval agent for targeted knowledge acquisition with a Chain-of-Thought reasoning agent to enable coherent, multi-step inference. This integrated design unifies retrieval-augmented generation with structured reasoning, leading to a 16.9% improvement in prediction accuracy over conventional LLM baselines. This work highlights the potential of combining cloud-based big data processing and collaborative agent architectures to advance personalized prediction systems, and we believe that this study could offer valuable insights to improve the design of personalized prediction systems through collaborative agent-based architectures. © 2025 IEEE.
Number of references: 0
Main heading: Forecasting
Controlled terms: Architecture - Autonomous agents - Big data - Cloud computing architecture - Computer architecture - Intelligent agents - Knowledge management - Multi agent systems
Uncontrolled terms: Agent architectures - Collaborative agents - Collaborative dual agent architecture - Heterogeneous characteristic - Language model - Multi dimensional - Multi-dimensional heterogeneous characteristic - Personalized prediction - RAG-CoT integration - Taste preference prediction
Classification code: 409.1 Architecture - 1101 Artificial Intelligence - 1104 Computer Architecture - 1105.1 Cloud and Edge Computing - 1106.2 Data Handling and Data Processing - 1106.5 Computer Applications - 1106.9 Computer Software - 1202 Statistical Methods
DOI: 10.1109/CBASE67452.2025.11335473
Compendex references: YES
Database: Compendex
Data Provider: Engineering Village